

Introduction to Abstract Algebra III: Homework 9

Due on June 5, 2026 at 23:59

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Exercise 14.4. Theorem 14.24 states that every p -group has nontrivial center. Find the center of D_8 .

Solution. Recall that D_8 has order 16 and is generated by a rotation ρ of order 8 and a reflection μ with $\mu\rho\mu^{-1} = \rho^{-1}$. An element is in $Z(D_8)$ if and only if it commutes with both ρ and μ .

For powers of ρ : every ρ^k commutes with ρ . We check which commute with μ : $\mu\rho^k = \rho^{-k}\mu$, so $\rho^k\mu = \mu\rho^k$ if and only if $\rho^k = \rho^{-k}$, i.e., $\rho^{2k} = e$, i.e., $8 \mid 2k$, i.e., $4 \mid k$. So among rotations, only $\rho^0 = e$ and ρ^4 are central.

For elements of the form $\mu\rho^k$: we check $(\mu\rho^k)\rho = \mu\rho^{k+1}$ and $\rho(\mu\rho^k) = \rho\mu\rho^k = \mu\rho^{-1}\rho^k = \mu\rho^{k-1}$. These are equal only if $\rho^{k+1} = \rho^{k-1}$, i.e., $\rho^2 = e$, which is false. So no reflection is central.

Therefore $Z(D_8) = \{e, \rho^4\} \cong \mathbb{Z}_2$. □

Exercise 14.5. Find the center of D_7 .

Solution. The group D_7 has order 14 and is generated by a rotation ρ of order 7 and a reflection μ with $\mu\rho\mu^{-1} = \rho^{-1}$.

For rotations ρ^k : $\mu\rho^k = \rho^{-k}\mu$, so ρ^k commutes with μ if and only if $\rho^k = \rho^{-k}$, i.e., $\rho^{2k} = e$, i.e., $7 \mid 2k$. Since $\gcd(2, 7) = 1$, this requires $7 \mid k$, so only $k = 0$, i.e., only e .

For reflections $\mu\rho^k$: as in the previous problem, $(\mu\rho^k)\rho = \mu\rho^{k+1}$ while $\rho(\mu\rho^k) = \mu\rho^{k-1}$, which are unequal since $\rho \neq e$. So no reflection is central.

Therefore $Z(D_7) = \{e\}$, which is consistent with D_7 not being a p -group (since $|D_7| = 14$ is not a prime power). □

Exercise 14.6. Let $G = X = S_3$ and make X a G -set using conjugation. That is, $\ast(\sigma, \tau) = \sigma\tau\sigma^{-1}$. Find all the orbits of X using this action. (Write permutations in disjoint cycle notation.)

Solution. The orbits under conjugation are precisely the conjugacy classes of S_3 . Elements of S_3 in disjoint cycle notation are:

$$S_3 = \{\iota, (1\ 2\ 3), (1\ 3\ 2), (1\ 2), (1\ 3), (2\ 3)\}.$$

Conjugate permutations have the same cycle type, so the conjugacy classes are determined by cycle type:

- Identity: $\{\iota\}$.
- 3-cycles: $\{(1\ 2\ 3), (1\ 3\ 2)\}$.
- Transpositions: $\{(1\ 2), (1\ 3), (2\ 3)\}$.

These three sets are the orbits of X under conjugation by G . □

Exercise 14.7. Let $G = D_4$ and X be the set of all subgroups of D_4 with order two. The set X is a G -set using conjugation, $\ast(\sigma, H) = \sigma H \sigma^{-1}$. Find all the orbits of this group action.

Solution. Using standard notation, $D_4 = \{\iota, \rho, \rho^2, \rho^3, \mu, \mu\rho, \mu\rho^2, \mu\rho^3\}$ where ρ is rotation by $\pi/2$ and μ is a reflection. The subgroups of order two are exactly the subgroups generated by elements of order two:

$$X = \{\langle \rho^2 \rangle, \langle \mu \rangle, \langle \mu\rho^2 \rangle, \langle \mu\rho \rangle, \langle \mu\rho^3 \rangle\}.$$

The subgroup $\langle \rho^2 \rangle = \{e, \rho^2\}$ is normal in D_4 since ρ^2 is central, so its conjugacy class is $\{\langle \rho^2 \rangle\}$.

For the reflections: μ and $\mu\rho^2$ are reflections across the diagonals, while $\mu\rho$ and $\mu\rho^3$ are reflections across the lines joining midpoints of opposite sides. Conjugating by ρ sends $\mu \mapsto \rho\mu\rho^{-1} = \mu\rho^2$, and $\mu\rho \mapsto \rho(\mu\rho)\rho^{-1} = \mu\rho^3$, so:

- $\{\langle \mu \rangle, \langle \mu\rho^2 \rangle\}$ is one orbit.
- $\{\langle \mu\rho \rangle, \langle \mu\rho^3 \rangle\}$ is one orbit.

Therefore the orbits are $\{\langle \rho^2 \rangle\}$, $\{\langle \mu \rangle, \langle \mu \rho^2 \rangle\}$, and $\{\langle \mu \rho \rangle, \langle \mu \rho^3 \rangle\}$. $\square$

Exercise 14.8. Let $G = U = \{z \in \mathbb{C} \mid |z| = 1\}$ be the circle group. Then $X = \mathbb{C}$, the set of complex numbers, is a G -set with group action given by complex number multiplication. That is, if $z \in U$ and $w \in \mathbb{C}$, $*(z, w) = zw$. Find all the orbits of this action. Also, find X_G .

Solution. For $w \in \mathbb{C}$, the orbit of w is

$$Gw = \{zw \mid z \in U\} = \{zw \mid |z| = 1\}.$$

Since $|zw| = |z||w| = |w|$, every element of Gw has modulus $|w|$. Conversely, if $w' \in \mathbb{C}$ with $|w'| = |w|$, then $z = w'/w \in U$ (when $w \neq 0$) satisfies $zw = w'$. Thus:

- If $w \neq 0$, the orbit of w is the circle $\{w' \in \mathbb{C} \mid |w'| = |w|\}$ of radius $|w|$ centered at the origin.
- If $w = 0$, the orbit is $\{0\}$.

So the orbits are $\{0\}$ and the circles centered at the origin of each positive radius $r > 0$.

For X_G , an element $w \in \mathbb{C}$ satisfies $zw = w$ for all $z \in U$ if and only if $(z - 1)w = 0$ for all $z \in U$, which (taking $z \neq 1$) forces $w = 0$. Hence $X_G = \{0\}$. $\square$

Exercise 14.10. Let G be a group of order 9 and suppose that $|X| = 10$. For each possible action of G on X , give a list of the orbit sizes. List the orbit sizes from largest to smallest.

Solution. By Theorem 14.17, each orbit size divides $|G| = 9$, so orbit sizes lie in $\{1, 3, 9\}$, and they must sum to $|X| = 10$. The possible partitions of 10 into parts from $\{1, 3, 9\}$ are:

- 9, 1
- 3, 3, 3, 1
- 3, 3, 1, 1, 1, 1
- 3, 1, 1, 1, 1, 1, 1, 1
- 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

Each of these is realizable, and these are all possible orbit size lists. $\square$

Exercise 14.19. Let X be a G -set. Show that G acts faithfully on X if and only if no two distinct elements of G have the same action on each element of X .

Solution. Recall that G acts faithfully on X means the kernel of the homomorphism $\varphi : G \rightarrow S_X$ defined by $\varphi(g) = \sigma_g$ is trivial, i.e., the only element of G fixing every $x \in X$ is e .

Suppose G acts faithfully and let $g_1, g_2 \in G$ have the same action on every $x \in X$, i.e., $g_1x = g_2x$ for all $x \in X$. Then $g_2^{-1}g_1x = x$ for all $x \in X$, so $g_2^{-1}g_1$ fixes every element of X . By faithfulness, $g_2^{-1}g_1 = e$, hence $g_1 = g_2$.

Conversely, suppose no two distinct elements of G have the same action on every $x \in X$. Let $g \in G$ fix every $x \in X$, so $gx = ex$ for all $x \in X$. By hypothesis, $g = e$. Hence the kernel of φ is trivial, so G acts faithfully. $\square$

Exercise 14.22. Let $\{X_i \mid i \in I\}$ be a disjoint collection of sets, so $x_i \cap X_j = \emptyset$ for $i \neq j$. Let each X_i be a G -set for the same group G .

- (i) Show that $\bigcup_{i \in I} X_i$ can be viewed in a natural way as a G -set, the union of the G -sets X_i .
- (ii) Show that every G -set X is the union of its orbits.

Solution to (i). Let $X = \bigcup_{i \in I} X_i$. Since the X_i are pairwise disjoint, every $x \in X$ belongs to a unique X_i . Define the action of G on X by $gx = g \cdot_i x$, where $\cdot_i$ denotes the given action of G on X_i and i is the unique index with $x \in X_i$. We verify the two conditions of Definition 14.3:

- (i) For any $x \in X_i$, $ex = e \cdot_i x = x$ since X_i is a G -set.
- (ii) For $g_1, g_2 \in G$ and $x \in X_i$, we have $g_2x = g_2 \cdot_i x \in X_i$ (since X_i is a G -set), and then $(g_1g_2)x = (g_1g_2) \cdot_i x = g_1 \cdot_i (g_2 \cdot_i x) = g_1(g_2x)$.

Hence X is a G -set. □

Solution to (ii). Let X be a G -set. By Theorem 14.15, the relation $x_1 \sim x_2$ if and only if $gx_1 = x_2$ for some $g \in G$ is an equivalence relation on X , whose equivalence classes are the orbits. Since equivalence classes partition X , every element of X belongs to exactly one orbit, and X is the union of its orbits. □

Exercise 14.29. Prove that if G is a group of order p^3 , where p is a prime number, then $|Z(G)|$ is either p or p^3 . Give an example where $|Z(G)| = p$ and an example where $|Z(G)| = p^3$.

Solution. Since $|G| = p^3$, G is a p -group, so by Theorem 14.24 the center $Z(G)$ is nontrivial. By Lagrange's theorem, $|Z(G)| \in \{p, p^2, p^3\}$. Suppose for contradiction that $|Z(G)| = p^2$. Then $|G/Z(G)| = p$, so $G/Z(G)$ is cyclic. But if $G/Z(G)$ is cyclic, then G is abelian, forcing $Z(G) = G$ and $|Z(G)| = p^3$, a contradiction. Hence $|Z(G)| \neq p^2$, and so $|Z(G)|$ is either p or p^3 .

Now, for the examples. The abelian group $G = \mathbb{Z}_{p^3}$ satisfies $Z(G) = G$, so $|Z(G)| = p^3$. For $p = 2$, the dihedral group D_4 has order $8 = 2^3$ and center $Z(D_4) = \{e, \rho^2\} \cong \mathbb{Z}_2$, so $|Z(D_4)| = 2 = p$. For odd p , the Heisenberg group modulo p ,

$$G = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \mid a, b, c \in \mathbb{Z}_p \right\},$$

has order p^3 and center $Z(G) \cong \mathbb{Z}_p$, giving $|Z(G)| = p$. □

Exercise 16.2. Let $\varphi : \mathbb{Z}_{18} \rightarrow \mathbb{Z}_{12}$ be the homomorphism where $\varphi(1) = 10$.

- (i) Find the kernel K of φ .
- (ii) List the cosets in $\mathbb{Z}_{18}/K$, showing the elements in each coset.
- (iii) Find the group $\varphi[\mathbb{Z}_{18}]$.
- (iv) Give the correspondence between $\mathbb{Z}_{18}/K$ and $\varphi[\mathbb{Z}_{18}]$ given by the map μ described in Theorem 16.2.

Solution to (i). Since φ is a homomorphism with $\varphi(1) = 10$, we have $\varphi(n) = 10n \pmod{12}$. The kernel is

$$K = \{n \in \mathbb{Z}_{18} \mid 10n \equiv 0 \pmod{12}\}.$$

We need $12 \mid 10n$, i.e., $6 \mid 5n$. Since $\gcd(5, 6) = 1$, this requires $6 \mid n$. The elements of $\mathbb{Z}_{18}$ divisible by 6 are $\{0, 6, 12\}$, so $K = \{0, 6, 12\}$. □

Solution to (ii). Since $|K| = 3$, the quotient $\mathbb{Z}_{18}/K$ has $18/3 = 6$ cosets:

$$\begin{aligned} 0 + K &= \{0, 6, 12\}, \\ 1 + K &= \{1, 7, 13\}, \\ 2 + K &= \{2, 8, 14\}, \\ 3 + K &= \{3, 9, 15\}, \\ 4 + K &= \{4, 10, 16\}, \\ 5 + K &= \{5, 11, 17\}. \end{aligned}$$
□

Solution to (iii). The image is $\varphi[\mathbb{Z}_{18}] = \{10n \pmod{12} \mid n \in \mathbb{Z}_{18}\}$. Since $\gcd(10, 12) = 2$, the image is the unique subgroup of $\mathbb{Z}_{12}$ of order $12/\gcd(10, 12) = 6$, namely $\varphi[\mathbb{Z}_{18}] = \langle 2 \rangle = \{0, 2, 4, 6, 8, 10\} \leq \mathbb{Z}_{12}$. $\square$

Solution to (iv). By the First Isomorphism Theorem, the isomorphism $\mu : \mathbb{Z}_{18}/K \rightarrow \varphi[\mathbb{Z}_{18}]$ is given by $\mu(n + K) = \varphi(n) = 10n \pmod{12}$:

$$\mu(0 + K) = 0, \quad \mu(1 + K) = 10, \quad \mu(2 + K) = 8, \quad \mu(3 + K) = 6, \quad \mu(4 + K) = 4, \quad \mu(5 + K) = 2. \quad \square$$